

Arpan Jain

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HackerRank | CodeChef | Codeforces

EDUCATION

B.Tech in Computer Science, KIET Group of Institution (Affil. to AKTU) Expected Jun 2026
CGPA: 8.6/10.0

Experience

Research Intern, IIT Delhi May 2025 – Aug 2025

- Developing a multimodal search engine that retrieves contextually relevant documents and ancient map content based on user queries using LLMs, embeddings, and natural language processing.
- Extracting text from ancient maps by building a hybrid pipeline combining OCR, computer vision, and LLMs to enhance text accuracy and support semantic search.

RESEARCH PAPERS

- AgroCARE: IoT-Based Crop Recommendation and Automated Irrigation** – ACET 2025
- SARCH: Multimodal Search for Archaeological Archives** – CIKM 2025

PROJECTS

AgroCARE July 2024 - Present

Created AgroCare, an IoT-driven smart farming system that enables real-time soil monitoring, automated irrigation, and crop recommendations using machine learning, optimizing resource efficiency and promoting sustainable agricultural practices.

Team Size: 4

My Role: Designed and deployed the AgroCare website.

GitHub: github.com/Agrocare

Sign to Speech Converter(Hand Speak) May 2024 - Jun 2024

Engineered a system that converts sign language gestures into audible speech for hearing impaired people using Python, OpenCV, and machine learning. Achieved 95% precision in gesture recognition and integrated real-time speech synthesis, improving communication efficiency by 80% and reducing delay by 30%.

Team Size: 4

My Role: Built a user-centric, interactive frontend for the website.

GitHub: github.com/Hand-Speak

SKILLS

Languages: Java, C++, Python, SQL

ML & AI: NLP, Computer Vision, Information Retrieval, LLMs, Embeddings

Frameworks: PyTorch, TensorFlow, HuggingFace

Web & Databases: React, HTML, CSS, JS

Tools & Cloud: Git, GitHub, Docker, VS Code

Skills: Data Structures & Algorithms, OOP, DBMS, OS

EXTRA-CURRICULAR ACTIVITIES

- Selected for the LinkedIn Learning Campus Connect Program; authored a Medium article sharing insights from the experience, and actively contributed to professional community growth through content creation.
- Led teams in national-level Hackathons including Smart India Hackathon (SIH) and InnoTech 2024, securing a win at InnoTech 2024 by driving innovation and team collaboration.